predictions are instruments to validate hypotheses. The objective of scientific research is not just to arrive at predictions: it is to understand how the world functions; to construct and develop an image of the world, a conceptual structure to enable us to think about it. Before being technical, science is visionary.

The verifiable predictions are the sharpened tool which allows us to find out when we have misunderstood something. A theory lacking empirical confirmation is a theory which has not yet passed its exams. Exams never end, and a theory is not completely confirmed by one, two or three experiments. But it progressively acquires credibility, stage by stage, as its predictions are revealed to be correct. Theories such as general relativity and quantum mechanics, which initially left many perplexed, earned their credibility gradually, as all of their predictions – even the most bizarre – were gradually confirmed by experiments and observations.

The importance of experimental proof, on the other hand, does not mean that, without new experimental data, we cannot make advances. It is often said that science takes steps forward only when there is new experimental data. If this were true, we would have little hope of finding the theory of quantum gravity before measuring something new, but this is patently not the case. Which new data were available to Copernicus? None. He had the same data as Ptolemy. Which new data did Newton have? Almost none. His real ingredients were Kepler's laws and Galileo's results. What new data did Einstein have to discover general relativity? None. His ingredients were special relativity and Newton's theory. It simply isn't true that physics advances only when it is afforded new data.

What Copernicus, Newton, Einstein and many others did was to build upon pre-existing theories which synthesized empirical knowledge across vast fields of nature, and to find a way of combining and rethinking them to improve the general picture.

This is the basis on which the best research on quantum gravity operates. The origin of knowledge, as always in science, is ultimately empirical. But the data on which quantum gravity is built is not new experiments: it is the theoretical edifices which have already structured our knowledge of the world, in forms which are only partly coherent. The 'experimental data' for quantum gravity are general relativity and quantum mechanics. Building on these, trying to understand how a world in which both quanta and curved space exist may be made coherent, we attempt to look towards the unknown.

The enormous success of the giants who have preceded us in similar situations, such as Newton, Einstein and Dirac, gives us encouragement. We do not presume to be of their stature. But we have the advantage of sitting on their shoulders, and this allows us to look further than they did. One way or another, we cannot but try.